Speaker Information

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Talk Information

Neural mechanisms for skilled motor control

Dexterous hand movements are highly developed in primates and indispensable for daily activities. Traditionally, such movements were thought to depend primarily on the evolutionarily “newer” direct corticomotoneuronal (CM) pathway. However, recent evidences suggest that the “older” indirect pathways, mediated by spinal premotor interneurons (PreM-INs), also play an important role, highlighting the need to clarify their functional differences. To address this, we recorded neuronal activity from spinal PreM-INs and CM cells in macaques during a precision grip task, and compared their roles in generating hand muscle activity. Our results show that PreM-INs exert stronger facilitation across a broader set of muscles, promoting synergistic coactivation, whereas CM cells provide more selective facilitation, enabling control of relatively individual muscles. Further decomposition analysis revealed that these two systems correspond to different control modes—synergy-based and individual-based control—balancing stability and flexibility. These findings redefine our understanding of primate dexterous hand control as emerging from the cooperative integration of evolutionarily distinct premotoneuronal systems.

Research Overview (for collaboration discussion)

[Research Interests] Neural and computational basis for skilled motor control

[Current Research Projects] Frontoparietal dynamics underlying voluntary reaching/ Neural network modelling of frontoparietal interactions/ Establishment of artificial proprioceptive for voluntary motor control

[Techniques] Electrophysiology (single unit recordings/ electrocorticography), Cortical cooling/ Neural network modeling/ Theoretical modeling

[Aspects or techniques of interest for future collaboration] Integration of large-scale neural recordings with advanced analysis of neural population dynamics/ Cross-species comparison of motor control and motor learning mechanisms

Short Bio

Ph.D. at Kyoto University/ Section Chief at the National Institute of Neuroscience, Postdoctoral Fellow at the Queen’s University/ Program-specific Associate Professor at Kyoto University/ Associate Professor at Tamagawa University.